

1.5 Trigonometry

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use exact unit-circle values and signs by quadrant. Identities $\sin^2 x + \cos^2 x = 1$ and $\tan x = \sin x / \cos x$ simplify expressions. For trig equations over an interval, find all solutions using symmetry and periodicity; do not accept inverse-calculator principal value alone.

Worked example: follow the reasoning

$\sin x = \frac{1}{2}$ on $0 \leq x < 2\pi$ gives $x = \pi/6$ or $5\pi/6$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $\cos x = -1/2$ for $0 \leq x < 2\pi$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Reference angle $\pi/3$ [1]; cosine negative in quadrants II and III [1]; $x=2\pi/3, 4\pi/3$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.